Laboratory Exercise 1: Asymptotic Notation: Digit Factorial Sum

Learning Outcomes:

- Implement digit sum factorial for an n-digit integer
- Plot the running time for a certain set of inputs
- Analyze the asymptotic notation of the implemented algorithm.

Procedure

1. Implement an algorithm the digit factorial sum of an integer e.g. 123 = 1! + 2! + 3!= 9. Preferably use C/C++ programming to implement your algorithm.
2. Place your algorithm on a time logging function to record the time for various set size of random inputs with your preferred size of input interval. Record in a file (.txt) the time for various sizes on inputs
3. Use Python programming in preferably in Jupyter notebook to plot the graph of the running time in which the size of input at the x-axis and time on the y-axis.
4. Count the primitives of your algorithm to get its asymptotic approximation of its running time.
5. Plot this equation on the graph that you've created to compare it with the actual running time.
6. Indicate the asymptotic notation of your algorithm.
7. Submit all your source codes (in compressed format) with a prefix of your family name (e.g. Regis_exer1.rar).